

INTRODUCTION

2.1 Background

Radiation monitoring is a key field of safety and protection in nuclear power, the environment, medical treatment, in industry, and in emergency response. The traditional method of radiation monitoring has been to either measure radiation at a set location or use handheld equipment to view radiation from a distance. In most cases, it is hard to visualize the radiation distribution over a large geographical area in real-time using the traditional radiation monitoring methods. These traditional approaches do not have the ability to precisely locate radioactive sources or offer much assistance to quick response in the event of a radiological emergency.

Intelligent radiation mapping systems are becoming more and more significant, because of the powerful development of **embedded systems, GPS technology, Internet of Things (IoT) and geospatial visualization** techniques. Integrating radiation detectors with GPS modules and edge computing platforms such as Raspberry Pi 5 enables the collection of radiation measurements together with precise geographical coordinates. These datasets can be used to create interactive heat maps, which will help you gain insights into the distribution of radiation in an area being monitored.

To overcome these challenges, the **RADMAP (RADiation MAPing)** project was developed to develop a prototype portable GPS supported radiation mapping system that can record real-time radiation data, environmental parameters and geographical information. The system connects hardware sensors and software visualization tools to ensure the accurate, scalable, and intelligent monitoring of radiation, which can be applied to nuclear safety in the future. By Integration all the other components we need to test and validate each and every particular phase of working of our device to detect gather the data and other functionality.

2.2 Project Motivation

The main motivation for this project was kind of to get past the limits of normal 2D radiation heatmaps. Sure, heatmaps do show the radiation intensity across different spots but they do

not really tell you where the radioactive source is in physical space. Finding the exact source, that is, the origin of radiation is one of the biggest challenges that show up in radiation monitoring systems.

To tackle this research issue , the project basically aimed at pushing conventional 2D radiation mapping into a 3D Digital Twin kind of setup, so it can handle future work on radioactive source localization. The motivation got even stronger because we needed to stitch together real time GPS coordinates, some environmental sensing data, camera feedback, and radiation measurements into one single visualization platform, even if parts of it feel kind of messy at first.

The internship also sort of gave me a chance to learn in a practical way embedded system development, plus Raspberry Pi programming , and GPS telemetry, radiation instrumentation, data visualization, backend development, and even research driven engineering. The whole project, it mixes software creation with real world nuclear technology use cases, which kind of helps move things forward for more intelligent radiation monitoring and environmental surveillance.

2.3 Project Scope

So the scope of the **RADMAP** project kind of includes the whole development of a GPS enabled radiation mapping setup using a **Raspberry Pi-5** and several integrated hardware parts. Basically, the project went through building structured radiation datasets, then making interactive **2D radiation heatmaps**, and at the same time plugging in the **GlobalSat BU-353N** GPS Receiver so it can capture real-time latitude longitude temperature humidity plus a timestamp. After that, all the gathered info got synced with the Raspberry Pi-5 which i guess ties everything together.

The project also covers the integration of a **Gamma Spectrometer (CNSPEC MCA)** for radiation sensing and also some validation, using familiar radioactive references like **Cobalt-60 (Co-60)**, and **Cesium-137 (Cs-137)**. From the radiation readouts that were collected, dynamic heatmaps showing radiation counts, peak intensity and even the spatial pattern were successfully made.

Another big part of this project, is the research and development of a 3D Digital Twin meant for future radioactive source localization. The whole system brings together live camera feedback, **GPS telemetry**, and radiation measurements so it can show a real time three dimensional view. Also, we built a backend architecture that kind a coordinates sensor talking, the environmental context data, the radiation processing, and then the visualization modules. Beside this, a kind of night vision style color grading method was also implemented, to show different radiation intensity levels directly on the live camera feed, which improves situational awareness when people do radiation monitoring.

The outcomes of this project really lay down a solid base for later upgrades, like AI-assisted source localization, plus **LiDAR SLAM integration**, and autonomous robotic navigation. There's also drone-based radiation mapping, and a cloud powered radiation monitoring system, for nuclear safety use-cases. In other words the whole thing becomes a stepping stone, for more advanced and maybe a bit more adaptive monitoring setups later on.

2.4 My Role in the Project

During the internship I sort of actively contributed to both the research and development stages of the RADMAP project, like it was not only one thing. I was involved in developing radiation datasets and getting them published on **Kaggle**. I also worked on designing interactive **2D radiation heatmaps**, and then I integrated the **GlobalSat BU-353N GPS receiver BU-353N** with a **Raspberry Pi-5** for real time telemetry acquisition. Besides that I processed environmental parameters such as latitude longitude temperature and humidity, you know, the usual signals that matter.

I also worked on the integration a validating of the Gamma Spectrometer, using Co-60, and Cs-137 radioactive sources, and while doing that I helped with verification routines. I developed the backend architecture for data acquisition and visualization a bit of it at least, then I researched and implemented the idea of a 3D digital twin for radiation source localization. On top of that, I designed a real-time color grading system, inspired by night-vision imaging and created a 3D printable prototype model of the RADMAP system as well. Overall, these contributions improved the prototype functionality a lot, and they supported its approval for more funding, plus deeper research work coming next.